

MT4214 - Algebraic Geometry

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Chapter 1

Motivation - Cayley-Hamilton theorem

Statement: Every Square Matrix over a commutative ring satisfies its own Characteristic Polynomial.

Proof. Step 1: Let A be a diagonal matrix with $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$ as the diagonal elements. Trivially, we can show that the Characteristic Polynomial will be evaluated as follows:

$$\begin{aligned}\chi_A(x) &= \det(A - xI_n) \\ &= (\lambda_1 - x)(\lambda_2 - x) \dots (\lambda_n - x) \\ &= 0\end{aligned}$$

Step 2: A is diagonalizable. Then there exists matrices B, D such that $A = BDB^{-1}$. A property that will be used is as follows: $\chi_A(x) = \chi_D(x)$. Now, if we calculate the Characteristic Polynomial for A :

$$\begin{aligned}\chi_A(A) &= \det(A - xI_n) \\ &= \chi_D(A) \\ &= B\chi_D(D)B^{-1} \\ &= 0\end{aligned}$$

Step 3: General A . We know that diagonalizable matrices are dense in $M_{n \times n}(\mathbb{C})$. Consider the following function:

$$\phi : M_{n \times n}(\mathbb{C}) \rightarrow M_{n \times n}(\mathbb{C})$$

such that $\phi(A) = \chi_A(A) = 0 \forall A \in \text{Diagonal Matrices}$. The above function is a continuous function [Trust me bro]. Now, $\{0\}$ is a closed set. Therefore the pullback of a closed set will have to be a closed set as well. But diagonal elements are dense in $M_{n \times n}(\mathbb{C})$. Therefore we use this to extend this to the entire topological space, $M_{n \times n}(\mathbb{C})$.

$$\phi(A) = 0 \forall A$$

□

But this above argument only for fields which are Cauchy Complete. What about the characteristic p fields. There is no obvious topology, and hence no dense set.

1.1 Zariski Topology on k^n

Let k be an algebraically closed field. We want to define a topology on k^n .

Define a ring $A = k[X_1, X_2, \dots, X_n]$ is the ring of polynomial in n variables. Now, choose an element $f \in A$.

$$f : k^n \rightarrow k \text{ where } (a_1, a_2, \dots, a_n) \rightarrow f(a_1, a_2, \dots, a_n)$$

Now, we define a set function as follows:

$$Z(f) = \{(a_1, a_2, \dots, a_n) \mid f(a_1, a_2, \dots, a_n) = 0\}$$

Here, $Z(f)$ can be empty. Extending this to multiple functions:

$$Z(f_1, f_2, \dots, f_m) = \bigcap_{1 \leq i \leq m} Z(f_i)$$

Let $I \subseteq A$ be an ideal.

$$f(I) = \bigcap_{p \in I} Z(p)$$

Lemma: Let $I \subseteq A$ be an ideal of A generated as follows: $I = \langle f_1, f_2, \dots, f_n \rangle$. Then:

$$\begin{aligned} Z(I) &= Z(\langle f_1, f_2, \dots, f_n \rangle) \\ &= Z(\{f_1, f_2, \dots, f_n\}) \end{aligned}$$

Proof. We will prove this by showing that one set contains the other and vise-versa.

Part 1: $Z(\{f_1, f_2, \dots, f_n\}) \subseteq Z(I)$ as each of the f_i is always contained in the I .

Part 2: Consider an element $f \in I$. As the ideal I is generated by f_i , f can be written as a linear combination of f_i s, with coefficients in A :

$$f = \sum_{i=1}^n c_i f_i$$

where, all $c_i \in A$. Now, if $\bar{a} \in Z(\{f_1, f_2, \dots, f_n\})$, $f_i(\bar{a}) = 0$ for all i . Therefore:

$$\begin{aligned} f(\bar{a}) &= \sum_{i=1}^n c_i f_i(\bar{a}) \\ &= 0 \end{aligned}$$

Hence, $Z(\{f_1, f_2, \dots, f_n\}) \subseteq Z(I)$. Therefore, $Z(\{f_1, f_2, \dots, f_n\}) = Z(I)$. □

1.1.1 $Z(I)$ form a Topology on K^n

We define a topology on K^n by claiming that the closed sets in K^n are defined by the sets $Z(I)$. It satisfies the axioms of Topology as follows:

1. $\emptyset$ is in the topology: A is an ideal of itself. $Z(A) = \emptyset$.
2. k^n is in the topology: 0 polynomial also forms an ideal of A . $Z(0) = k^n$
3. Finite Union of $Z(I)$ belong to the topology: Let I, J, IJ be Ideals of A .

Proposition: $Z(I) \cup Z(J) = Z(IJ)$.

Proof. Consider an $a \in Z(I)$. This implies that $f(a) = 0 \forall f \in I$. Using that we can say that a is a solution for any polynomial of the form $f \cdot g$ where, $f \in I$ and $g \in A$. Now, similarly, consider a b in $Z(J)$. It would be a solution for any polynomial of the form $f \cdot g$ where, $f \in J$ and $g \in A$. Hence, $a \in Z(I) \cup Z(J) \Rightarrow a \in Z(IJ)$

Consider an $a \in Z(IJ)$. This implies that for all $f \in I$ and $g \in J$, the product $f(a)g(a) = 0$. As A is an integral domain, one of the factors must be 0. Let $a \notin I$. Then, there exists an $f \in I$ such that $f(a) \neq 0$. But, as $a \in Z(IJ)$, for every $g \in J$, $f(a)g(a) = 0$. This implies $g(a) = 0$ for every $g \in J$. Hence, $a \in Z(IJ) \Rightarrow Z(I) \cup Z(J)$ $\square$

4. Arbitrary Intersection of $Z(I)$ belong to the topology: let $I, J, I + J$ be Ideals of A .

Proposition: $Z(I) \cap Z(J) = Z(I + J)$.

Proof. Consider an $a \in Z(I) \cap Z(J)$. This implies that for any two polynomials $f \in I$ and $g \in J$, the sum $(f + g)(a) = 0$. Therefore, $a \in Z(I + J)$.

Now, let $a \in Z(I + J)$. Trivially, $a \in Z(I)$ and $a \in Z(J)$. $\square$

Therefore, $Z(I)$ forms a topology on K^n . $\mathbb{A}_k^n$ is defined as the vector space K^n with the Zariski Topology.

Exercise: Let X be a Compact Hausdorff Topological Space.

$$C(X) = \{f : X \rightarrow \mathbb{R} \mid f \text{ is continuous.}\}$$

Show that $C(X)$ is an $\mathbb{R}$ -Algebra on X and that it forms a Commutative Ring.

1.1.2 Category Theory

Definition: A Category C is a collection of objects $\text{ob}(C)$ and the following conditions:

1. For any two elements $x_1, y_1 \in \text{ob}(C)$, there is a set of morphisms $\text{Hom}_C(x_1, y_1)$. (We can think of $\text{ob}(C)$ as dots and $\text{Hom}_C(x_1, y_1)$ an arrow (could be bidirectional) from x_1 to y_1).

2. Composition of Morphisms is defined as follows: For any $x, y, z \in \text{ob}(C)$

$$\begin{aligned} \text{Hom}(x, y) \times \text{Hom}(y, z) &\rightarrow \text{Hom}(x, z) \\ f \times g &\mapsto g \circ f \end{aligned}$$

Which follow these axioms:

1. If $x, x', y, y' \in \text{ob}(C)$, and $x \neq x', y \neq y'$, then $f \in \text{Hom}(x, y)$ does not belong to $\text{Hom}(x', y')$.
2. For any $x, y, z \in \text{ob}(C)$, $x \circ (y \circ z) = (x \circ y) \circ z$.
3. For all $x \in \text{ob}(C)$, there exists identity map $I_x \in \text{Hom}(x, x)$ such that:

with the following properties:

1. If $x \neq x'$ or $y \neq y'$, then:

$$\text{Hom}_C(x, y) \cap \text{Hom}_C(x', y') = \emptyset$$

2. Composition is associative:

$$(f \circ (g \circ h)) = ((f \circ g) \circ h) = (f \circ g \circ h)$$

3. Identity maps: for every $x, y \in \text{ob}(C)$, there exist $I_x(x, x) \in \text{Hom}_C(x, x)$ and $I_y(y, y) \in \text{Hom}_C(y, y)$:

$$f \circ I_x = f \text{ and } I_y \circ f = f$$

And $I_x \circ f$ doesn't make any sense as the set maps don't align.

Example: Groups, Rings, k -Algebra, C_{opp} .
 C_{opp} is constructed as:

- Consider a Category C .
- C_{opp} contains the same points a C .
- $\text{Hom}_{C_{\text{opp}}}(y, x) = \text{Hom}_C(x, y)$

1.1.3 Commutative Algebra

Noetherian Ring: Let R be a commutative ring with unity. R is Noetherian if every ideal of R is finitely generated.

OR

R is Noetherian if every increasing sequence $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$ of ideals has a largest element.

Theorem: Show that the two definitions of a Noetherian Ring are equivalent. I.e., If R has Ascending Chain Condition $\Leftrightarrow$ All ideals of R are finitely generated.

Proof. The proof for this will be done in two parts.

Part 1: Consider a Commutative Ring R with a 1 and the Ascending Chain Condition. Let I be an Ideal which is not finitely generated. We can form the following chain:

$$\begin{aligned} I_0 &= 0 \\ I_1 &= \langle f_1 \rangle & f_1 &\in I \setminus I_0 \\ I_2 &= \langle f_1, f_2 \rangle & f_2 &\in I \setminus I_1 \\ &\vdots \\ I_n &= \langle f_1, f_2, \dots, f_n \rangle & f_n &\in I \setminus I_{n-1} \end{aligned}$$

$$I_0 \subsetneq I_1 \subsetneq I_2 \subsetneq \dots \subsetneq I_n \subsetneq \dots$$

Now, as R has Ascending Chain Condition, I_n stabilizes, i.e., there will be no f to pick from $I \setminus I_m$ for some finite m . Therefore, I is finitely generated.

Part 2: Consider a Commutative Ring R with a 1 and every ideal in R is finitely generated. Let there exist an Infinite Ascending Chain of Ideals, which does not stabilize:

$$I_0 \subsetneq I_1 \subsetneq I_2 \subsetneq \dots \subsetneq I_n \subsetneq \dots$$

As every ideal is finitely generated, consider the ideal:

$$I = \bigcup_{n=0}^{\infty} I_n$$

Let I be generated by $\{f_1, f_2, \dots, f_m\}$. As each $f_i \in I$, there exists a N_i such that $f_i \in I_{N_i}$. Let $N = \max(N_1, N_2, \dots, N_m)$. Then, for all i , $f_i \in I_N$. Therefore, $I \subseteq I_N$. But, by construction, $I_N \subseteq I$. Hence, $I = I_N$. But, as the chain is strictly increasing, there exists a I_{N+1} such that $I_N \subsetneq I_{N+1} \subseteq I$. This is a contradiction. Therefore, the chain stabilizes.

□

Hilbert Basis Theorem: Let R be a Noetherian Ring. Then, the Polynomial Ring $R[X]$ is also Noetherian.

Proof. Assume on the contrary that $R[X]$ is not Noetherian. Then, there exists a non-zero ideal $I \subseteq R[X]$ which is not finitely generated. Now, let $f_0 \in I$ be a polynomial of the smallest degree in I . Let, $I_0 = (f_0)$ be the smallest ideal containing f_0 . As I_0 does not generate I , there exists an element $f_1 \in I \setminus I_0$ of the smallest degree. Similarly, we can construct a sequence of polynomials $\{f_n\}$ such that:

$$\begin{aligned} f_n &\in I \setminus (f_0, f_1, \dots, f_{n-1}) \\ \deg(f_n) &\geq \deg(f_{n-1}) \end{aligned}$$

Let a_n be the leading coefficient of f_n . Now, consider the ideal $J = (a_0, a_1, a_2, \dots)$ in R . As R is Noetherian, J is finitely generated. Therefore, there exists a finite set of generators $\{a_0, a_1, \dots, a_m\}$ for J . This implies that for all $n > m$, there exists

$$a_n = \sum_{i=0}^m b_i a_i \text{ where } b_i \in R$$

Now, consider the polynomial:

$$g(X) = \sum_{i=0}^m b_i f_i(X)$$

The leading coefficient of $g(X)$ is a_n and $\deg(g) < \deg(f_n)$ as $\deg(f_i) \leq \deg(f_n)$ for all $i \leq m$. Now, consider the polynomial $h(X) = f_n(X) - g(X)$. The leading coefficient of $h(X)$ is 0 and $\deg(h) < \deg(f_n)$. But, as $f_n(X) \in I$ and $g(X) \in I$, $h(X) \in I$. This contradicts the fact that $f_n(X)$ is the polynomial of the smallest degree in $I \setminus (f_0, f_1, \dots, f_{n-1})$. Therefore, our assumption is wrong and $R[X]$ is Noetherian. $\square$

1.2 Affine Algebraic Space

Consider the set $\mathbb{A}_k^n$ with the Zariski Topology, where k is algebraically closed. Consider $Z(I)$. Let $Y \subseteq \mathbb{A}_k^n$ be a subset.

$$I(Y) = \{f \in k[X_1, X_2, \dots, X_n] \mid f(a) = 0 \text{ for all } a \in Y\}$$

Hilbert's Nullstellensatz Theorem: Let $\theta \in k[X_1, X_2, \dots, X_n]$.

$$I(Z(\theta)) = \sqrt{\theta}$$

where, $\sqrt{\theta}$ is a radical ideal, and requires k to be algebraically closed.

Corollary: There is a one-to-one correspondence between the radical ideals of $k[X_1, X_2, \dots, X_n]$ and the closed subsets of $\mathbb{A}_k^n$.

$$\{Y \subseteq \mathbb{A}_k^n \mid Y \text{ is closed}\} \leftrightarrow \{\sqrt{\theta} \subseteq k[X_1, X_2, \dots, X_n] \mid \sqrt{\theta} \text{ is a radical ideal}\}$$

Corollary: Maximal ideals of $k[X_1, X_2, \dots, X_n]$ are exactly the ideals $(X_1 - a_1, X_2 - a_2, \dots, X_n - a_n)$.

Proof.

$$k[X_1, X_2, \dots, X_n]/(X_1 - a_1, X_2 - a_2, \dots, X_n - a_n) \cong k$$

$\square$

Now, let $m \subseteq k[X_1, X_2, \dots, X_n]$ be a maximal ideal. Then, $I(Z(m)) = m$ by Nullstellensatz. This implies that, for every maximal ideal, $Z(m) = \emptyset$.

Let $(a_1, a_2, \dots, a_n) \in Z(m)$.